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Author(s): Ricardo Lagos

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# A Model of TFP

RICARDO LAGOS
*Federal Reserve Bank of Minneapolis and New York University*

This paper proposes an aggregative model of total factor productivity (TFP) in the spirit of Houthakker (1955–1956). It considers a frictional labour market where production units are subject to idiosyncratic shocks and jobs are created and destroyed as in Mortensen and Pissarides (1994). An aggregate production function is derived by aggregating across micro-production units in equilibrium. The level of TFP is explicitly shown to depend on the underlying distribution of shocks as well as on all the characteristics of the labour market as summarized by the job-destruction decision. The model is also used to study the effects of labour-market policies on the level of measured TFP.

## 1. INTRODUCTION

This paper focuses on the theory underlying the aggregate production function and shows how labour-market policies can affect this function in general and the level of measured total factor productivity (TFP) in particular. Specifically, I construct an aggregative model of TFP in the spirit of Houthakker (1955–1956): the basic idea is to derive an aggregate production function by aggregating across active production units. In equilibrium, the levels of output, inputs, and TFP, as well as the shape of the aggregate relationship between them, depend on individual production decisions—such as which production units remain active in the face of idiosyncratic shocks—and these decisions are, in turn, affected by policies. In this way, the model can be used to study the precise interaction between all these variables explicitly.

In the model proposed here, policy affects TFP because the latter is related to the average productivity of the units that are active, and policy induces changes in the productivity composition of active units. By distorting the way in which individual production units react to the economic environment, labour-market policies can make an economy exhibit a low level of TFP. As a result, two economies may exhibit different levels of TFP even if production units in both have access to the same technology and are subject to identical shocks.

At a theoretical level, the paper also shows that, under some conditions, a standard search model of the labour market—with its underlying meeting frictions and simple fixed-proportions micro-level production technologies—can generate an aggregate production function that looks just like the one implied by the textbook neoclassical model of growth in which firms have access to a standard constant-returns Cobb–Douglas production technology. So in this sense, from the perspective of aggregate output, inputs, and productivity, the neoclassical and the search paradigms can seem quite close. However, the search model implies a different mapping between the parameters of the aggregate production technology and observables, and this feature can be relevant for growth accounting exercises.

At a conceptual level, the paper is related to the vast literature that documents and tries to explain differences in TFP levels across countries. Examples include Klenow and Rodríguez-Clare (1997), Hall and Jones (1999), and Parente and Prescott (2000). In terms of explanations, this literature shares the basic idea that the level of an economy's TFP is determined by the quality of its "institutions". Hall and Jones (1999) argue that differences in observed TFP are driven by differences in the institutions and government policies they collectively refer to as "social infrastructure". Corrupt government officials, severe impediments to trade, poor contract



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enforcement, and government interference in production are some of their examples of bad social infrastructures that could lead to low levels of TFP.

Parente and Prescott (1994) propose that some countries have lower TFP than others because their process of technology adoption at the micro-level is constrained by "barriers to riches". These barriers are essentially any institution or government policy that increases the cost of technology adoption. From that perspective, this paper can be thought of as adding labour-market policies to the list of institutions that affect the level of TFP.[1]

The paper is also related to the large body of work that tries to account for differences in macroeconomic performance, usually employment rates and labour productivity, with differences in labour-market policies, typically unemployment benefits or employment protection. Hopenhayn and Rogerson (1993), Millard and Mortensen (1997), Marimon and Zilibotti (1999), Veracierto (2001), and Restuccia and Rogerson (2004) are some examples of this line of work.

The rest of the paper is organized as follows. Section 2 lays out the model. The equilibrium is characterized in Section 3. In Section 4, an aggregation result similar in spirit to the classic contribution of Houthakker (1955–1956) is derived for a canonical dynamic equilibrium search economy. This section also shows how, when aggregate inputs are measured appropriately, the level of TFP depends on all the characteristics of the labour market summarized by the job-destruction decisions. Section 5 introduces four policies—employment subsidies, hiring subsidies, firing taxes, and unemployment benefits—and studies their effects on TFP. Section 6 puts the theoretical results of the previous sections into perspective. Section 6.1 explains the similarities and differences between the aggregation result obtained in Section 4 and Houthakker's contribution and relates the general approach to the literature on aggregation of production functions that dates back to the late 1940's. Since empirically, the level of TFP is measured as a residual, it depends critically on the measurement of the inputs of production. Section 6.2 discusses some of these measurement issues and uses the theory to study their implications for the nature of the TFP residual. Section 6.3 draws some connections to the standard growth accounting exercises. In Section 7, a calibrated version of the model is used to quantify the effects of labour-market policies on the level of TFP. Section 8 extends the basic model to allow for serially correlated shocks and state-dependent destruction rates, and it generalizes the main aggregation result. Section 9 concludes. All propositions are proved in Appendix A. Appendix B contains additional extensions and some auxiliary results.[2]

## 2. THE MODEL

The labour market is modelled as in Mortensen and Pissarides (1994).[3] Time is continuous and the horizon infinite. There is a continuum of infinitely lived agents of two types: workers and

1. Unlike Hall and Jones (1999), Parente and Prescott (2000), or Acemoglu and Zilibotti (2001), this paper is not about development economics, in the sense that it does not attempt to explain why some countries are 30 times richer than others in *per capita* terms. The emphasis on labour-market institutions makes the analysis more relevant to study productivity differences among a relatively homogeneous set of countries (or sectors). For example, Hall and Jones (1999) and Klenow and Rodríguez-Clare (1997) report productivity differences among the Organisation for Economic Co-operation and Development (OECD) countries, with France and Italy having a higher level of TFP and Germany a much lower level relative to the U.S. These observations may seem striking, especially given the conventional wisdom that France and Italy have distorted labour markets vis-à-vis the U.S. These are examples of the types of questions that the model developed here is well suited to address.

2. Appendix B can be found in the working paper version (Lagos, 2006) or downloaded from the *Review*'s webpage for supplementary materials: http://www.restud.org.uk/supplements.htm.

3. There are at least three reasons for carrying out the analysis in a search and matching framework. First, as will be discussed in Section 5, the labour-market policies considered will have testable implications not only for the level of TFP but also for the unemployment rate and the job-creation and job-destruction rates. Second, an explicit treatment of unemployment is relevant because—as will be shown in Sections 6 and 7—the unemployment rate will affect

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firms. Both types are risk neutral. The size of the labour force is normalized to unity, whereas the number of firms will be determined endogenously by free entry. Workers derive utility from consumption, and in one of the specifications developed below, also suffer disutility from working. Each firm has a single job that can be either filled or vacant and searching. Similarly, workers can be either employed by a firm or unemployed and searching. No new offers arrive while an agent is in a relationship. I abstract from capital accumulation and assume an exogenous rental rate of capital, $c$. The aggregate stock of capital, $K$, will be determined by demand.[4]

Assume meeting frictions can be represented by a function $m(u, v)$ that determines the instantaneous number of meetings as a function of the numbers of searchers on each side of the market, namely, unemployed workers $u$ and vacancies $v$. Suppose $m$ exhibits constant returns to scale and is increasing in both arguments. Let $q(\theta)$ denote the (Poisson) rate at which a vacancy contacts an unemployed worker, with $\theta = v/u$.[5]

Each firm has access to a technology $f(x, n, k)$ that combines the hours supplied by the worker it employs, $n$, and capital, $k$, to produce a homogeneous consumption good. The match-specific productivity level is stochastic and indexed by the random variable $x$. I assume that

$$f(x, n, k) = x \min(n, k) \tag{1}$$

and interpret $k$ as the firm's capacity or scale of operation. Thus, output is linear in hours but is bounded above by the stock of capital the firm is operating with. The convention is that the technology is such that all projects have the same scale of operation $k$. Every firm has to rent and put in place $k$ units of capital to be able to engage in search while vacant and to produce while filled.[6] This captures the notion that hours are a fully flexible factor, but capital is relatively fixed. Firms rent capital from a competitive market at flow cost $c$.

The process that changes the match-specific productivity, $x$, is Poisson with finite arrival rate $\lambda$. When a match of productivity $x$ suffers a change, the new value $x'$ is a draw from the fixed distribution $G(\cdot)$. So the productivity process is persistent (since $\lambda < \infty$), but—conditional on change—it is independent of the firm's previous state.[7] The Poisson process and the productivity draws are independently and identically distributed across firms, and there is no aggregate uncertainty. The focus will be on steady-state outcomes.

In the next section, I will show that there is a unique productivity level $R_t$ such that active matches dissolve if productivity ever falls below that level and new matches form only if their initial productivity is at least $R_t$.[8] Let $H_t(x)$ denote the cross-sectional productivity distribution of active matches. That is, $H_t(x)$ is the fraction of matches producing at productivities $x$ or lower

empirical measures of TFP for the ways of measuring aggregate inputs that can be found in the literature, and finally, this framework has already been used extensively to analyse the effects of similar policies on many other aggregate labour-market outcomes (see Ljungqvist and Sargent, 2000; Pissarides, 2000; and references therein).

4. This is the usual small, open-economy assumption. The model abstracts from saving and accumulation because the focus here is on isolating the effects of labour-market policies on the level of TFP. But even in the context of trying to explain income differences, Prescott (1998) and Parente and Prescott (2000) conclude that one cannot rely on policies that cause differences in saving rates, as they do not vary systematically with countries' incomes.

5. Note that $q(\theta) = m(1/\theta, 1)$, and hence $q' < 0$. The rate at which a worker contacts a vacancy is $\theta q(\theta)$ and is increasing in $\theta$. For some of the existence proofs, assuming $\lim_{\theta \to 0} q(\theta) = \infty$ will also be useful.

6. The idea is that in order to search, the firm must have borrowed some capital, say, to set up a plant, and plants come in a single size, $k$.

7. This is the process used by Mortensen and Pissarides (1994). For reasons that will become clear shortly, Section 8 generalizes the model by specifying that when a match of productivity $x$ suffers a change, the new value $x'$ is a draw from the fixed distribution $G(\cdot \mid x)$. If $G(\cdot \mid x_1) < G(\cdot \mid x_0)$ when $x_0 < x_1$, then apart from being persistent, the idiosyncratic shocks are also positively correlated over time.

8. Mortensen and Pissarides (1994) work with a bounded support and assume new matches start off with the highest productivity. I relax these assumptions and treat active and new matches symmetrically. In the model considered here, the initial productivity of a match is a nondegenerate random variable drawn from the same distribution as the innovations to active matches.

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